Unit 15 - Probability	
Lesson	**Page Number**

The following B.E.S.T Standards will be covered in this unit:

MA.7.DP.2.1 Determine the sample space for a simple experiment.

- *Clarification 1: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.*

MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

- *Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal between 0 and 1 with probabilities close to 1 corresponding to highly likely events and probabilities close to 0 corresponding to highly unlikely events.*
- *Clarification 2: Instruction includes $P(event)$ notation.*
- *Clarification 3: Instruction includes representing probability as a fraction, percentage or decimal.*

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

- *Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal.*
- *Clarification 2: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.*

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

- *Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal.*
- *Clarification 2: Instruction includes recognizing that experimental probabilities may differ from theoretical probabilities due to random variation. As the number of repetitions increases experimental probabilities will typically better approximate the theoretical probabilities.*
- *Clarification 3: Experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.*

Unit 15, Lesson 1: Exploring Sample Space

Benchmark

MA.7.DP.2.1 Determine the sample space for a simple experiment.

Learning Targets

- ☐ I can determine all possible outcomes of a given scenario.
- ☐ I can determine whether or not a simple experiment is fair.

15.1.1 Warm-Up: Carolyn Beatrice Parker

Carolyn Beatrice Parker, born in Gainesville, FL in 1917, was the first African American woman known to receive a graduate degree in physics and subsequently work on the top-secret government project known as the Manhattan Project. Before earning her master's degree in physics, she earned a master's degree in mathematics and was also a teacher in Florida.

In 2020, an elementary school in Gainesville, FL was re-named in her honor. At this time, 11 of the 21 elementary schools in the Alachua County Public School system were named after people.

1. Without performing any calculations, select the statement that best describes the percentage of the elementary schools in Alachua County that were named after people as of 2020.

   ○ significantly less than 50%
   ○ slightly less than 50%
   ○ exactly 50%
   ○ slightly more than 50%
   ○ significantly more than 50%

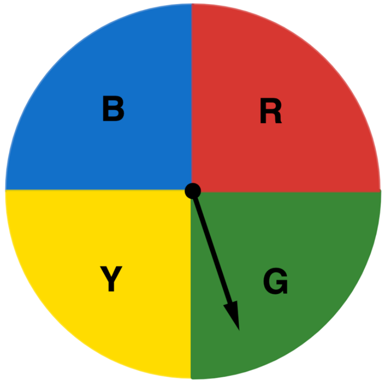
2. Explain your reasoning.

15.1.2 Guided Instruction: Determining Outcomes

An ***outcome*** is a possible result of an experiment.

1. A spinner divided into four equal areas is shown.

   a. How many outcomes are there from spinning the spinner?



In a probability model for a random process, the ***sample space*** is a list of the individual outcomes that are being considered.

   b. With your partner, discuss the sample space for the spinner. Write a list of the possible outcomes.

An ***event*** is a set of possible outcomes resulting from an experiment. In general, an event is any subset of a sample space.

   c. How many outcomes are included in the event that the spinner lands on blue?

   d. Primary colors include yellow, blue, and red. How many outcomes are included in the event that the spinner lands on a primary color?

2. Doug's parents created a set of cards with chores on them. Each Saturday, Doug chooses a card to determine his chore.

Mow the yard	Dust and vacuum the house	Clean the bathrooms	Clean your bedroom
Clean your bedroom	Mow the yard	Weed the garden	Mow the yard
Clean the bathrooms	Parents' choice	Dust and vacuum the house	Free day

a. What are the possible outcomes for the chore cards?

b. Which outcomes have more than one card?

c. On a particularly hot Saturday, Doug is hoping he doesn't select an outdoor chore. Which outcomes could be included in the event that he chooses an outdoor chore?

d. How many outcomes could be included in the event that Doug chooses an indoor chore?

3. Use the word bank to complete the summary of key concepts below.

event	outcomes	sample space	set

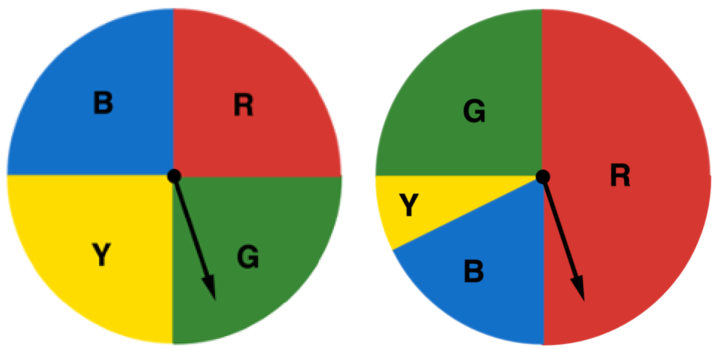
To determine the ________________ of a simple experiment, find all the ________________ that are possible and write them as a ________________.

An ________________ is a subset of the sample space, such as rolling an even number on a 6-sided die.

15.1.3 Exploration: Digging Deeper Into Outcomes

Two spinners are shown.

Spinner A **Spinner B**



1. Work with your partner to compare and contrast the features of the two spinners.

Similarities	Differences

2. One of the spinners is considered a fair spinner. Complete the sentence by selecting the fair spinner and writing an explanation.

○ Spinner A
○ Spinner B

is the fair spinner because ______________

____.

15.1.4 Activity: Gallery Walk

1. Work with your partner to determine the sample space for each game and whether the game is fair. Fill in the chart with your answers.

Game Name	Sample Space	Is it fair?
Skee-Ball		
Card Draw		
Prize Wheel		
Rubber Ducks		

2. Which game has the largest sample space?

3. Discuss with your partner which game's sample space was the most difficult to determine.

4. Carnivals include many games similar to these. What do you notice about the fairness of such games?

5. Find another pair of partners to form a group of four. Compare your responses on whether each game is fair and discuss your reasoning. Record one similarity and one difference you notice in your reasoning.

Similarity	Difference

6. Future lessons in this unit will explore probabilities with fair coins, fair dice, and fair spinners. Describe what you think it means for a coin, a die, or a spinner to be fair.

15.1.5 Wrap Up: Reflection

Complete each statement based on your understanding at this point.

1. I am still unsure about ______________________________

____

______________________________.

2. One question I have that will help me better understand

is ______________________________

____

______________________________?

Unit 15, Lesson 2: Determining Sample Space

Benchmark

MA.7.DP.2.1 Determine the sample space for a simple experiment.

Learning Target

☐ I can determine the sample space for a simple experiment.

15.2.1 Warm-Up: Which Game Would You Choose?

1. Two games are described.

   Game 1: A player flips a coin and wins if it lands on heads.

   Game 2: A player rolls a standard number cube and wins if it lands on a number that is divisible by 3.

   a. Which game would you rather play?

   b. Explain your reasoning.

15.2.2 Exploration: Simple Experiments

Work with your partner to complete the following.

1. Discuss the sample space for both games from the Warm-Up with your partner.

   a. Write the sample space for each in the table.

Random Process	Sample Space
Flipping a Coin 	
Rolling a Standard Number Cube 	

   b. What are the possible outcomes in the event of landing on heads in Game 1 from the Warm-Up?

   c. What fraction of the possible outcomes from Game 1 results in a win?

   d. What are the possible outcomes in the event of rolling a number divisible by 3 in Game 2 from the Warm-Up?

   e. What fraction of the possible outcomes from Game 2 results in a win?

   f. Based on your answers to Parts B-E, complete the statement.

   A player has a greater likelihood of winning

   ○ Game 1
   ○ Game 2

   because $\frac{\square}{\square} > \frac{\square}{\square}$.

   g. How could Game 2 be modified so the likelihood of winning either game is the same?

2. With your partner, determine who is partner A and who is partner B. For this activity, you and your partner will pull slips of paper from a bag.

   a. Make a guess about what is on the slips of paper.

   b. Without looking in the bag, partner A should take out one of the papers and show it to partner B. Both partners record what was printed on the paper in the first row of the table.

Who?	What is printed on the paper?
Partner A, Choice 1	
Partner B, Choice 1	
Partner A, Choice 2	
Partner B, Choice 2	
Partner A, Choice 3	
Partner B, Choice 3	

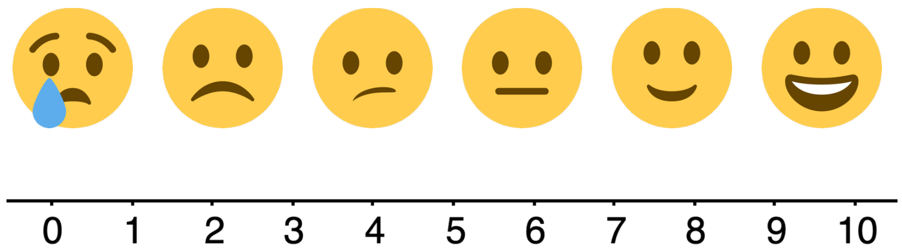
   c. Partner A should put the paper back into the bag. Then, partner B should take out one of the papers and show it to partner A for both to record in the table.

   d. After both partners have made their first selection and put it back in the bag, make a revised guess of the sample space.

   e. Continue taking turns and recording the results in the table until each partner has taken three turns.

   f. Make a revised guess of the sample space.

   g. Rate your confidence in your last guess using the emoji scale.



   h. Discuss with your partner what you could do to make a better guess of the sample space. Write at least two strategies you discussed.

   i. Look at all the papers in the bag. Write the sample space.

15.2.3 Bring It Together: Defining Simple Experiments

The games from the exploration are all considered **simple experiments**.

A ***simple experiment*** is an action that has a definite outcome that cannot be predicted with certainty.

In a simple experiment, each outcome in the sample space is equally likely to happen; however, events, which are subsets of the sample space, may have different likelihoods of happening. This is called probability.

The ***probability*** of an event is a measure of the likelihood the event will occur.

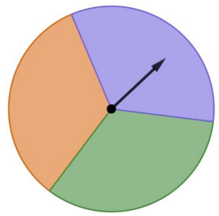
1. Consider the simple experiment involving pulling papers from the bag in the Exploration.

    a. Explain whether all of the possible outcomes in the sample space were equally likely or not.

    b. Explain which of the following events would have been more likely to happen in that experiment – pulling out a paper with a vowel on it or pulling out a paper with a consonant on it.

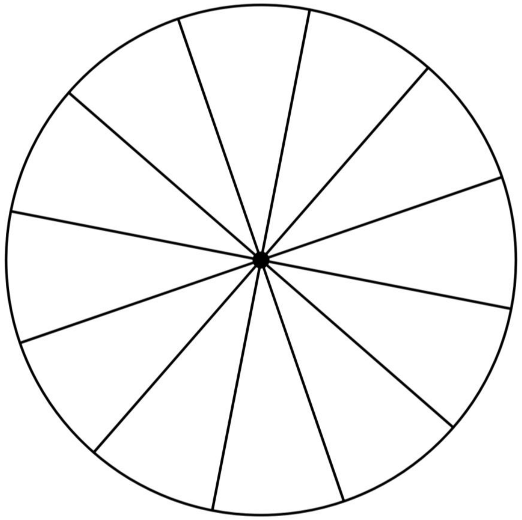
15.2.4 Guided Practice: Determining Sample Space of Simple Experiments

1. For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Spinning the spinner 	
Rolling a fair 12-sided die 	

2. Complete the following.

   a. Given the sample space of {Reese's, Snickers, Twix, Kit Kat}, label the game wheel so that there is a fair chance of winning each candy bar as a prize.



   b. Explain how you determined the labels for the game board.

15.2.5 Wrap-Up: Ticket Out the Door

1. For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Selecting a day of the week from a deck of cards with each day written on one card	
Flipping an integer chip 	

Unit 15, Lesson 3: Probability

Benchmark

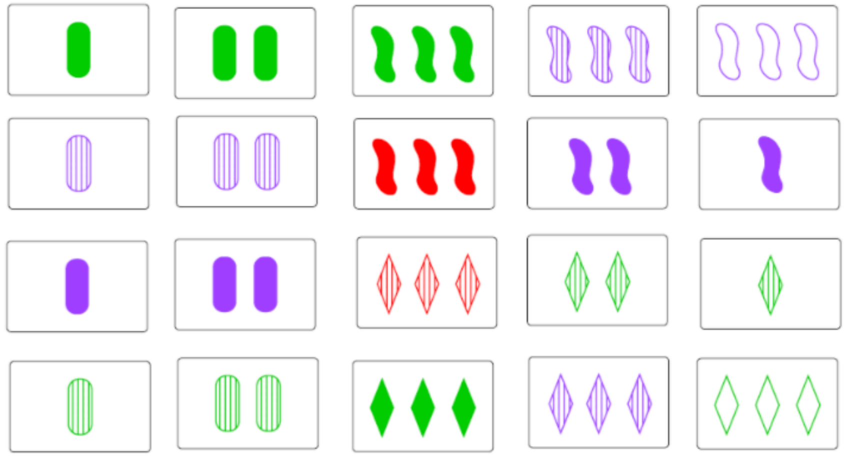
MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

Learning Targets

- ☐ I can express the probability of a chance event using a fraction, percentage, or decimal.
- ☐ I can compare probabilities of chance events.

15.3.1 Warm-Up: Notice and Wonder

1. Some of the cards in the card game SET are shown.



   a. What do you notice?

   b. What do you wonder?

15.3.2 Exploration: Representing Probabilities

When performing simple experiments, all of the outcomes have the same likelihood of happening. For example, when flipping a fair coin, the sample space includes two possible outcomes, {heads, tails}. Each outcome has an equal chance of happening. Therefore, the chance of landing on heads is $\frac{1}{2}$ because this event represents 1 outcome out of 2 possible outcomes. Represented as a decimal or a percentage, the chance of landing on heads is 0.5 or 50%, respectively.

Work with your partner to complete the following.

1. From the percentages listed below, determine the percent chance of each event occurring. Each value is used once.

0% $\quad 16.\overline{6}\%$ $\quad 33.\overline{3}\%$ $\quad 50\%$ $\quad 100\%$

Event	Percent Chance
a. When given three doors with only one containing a prize behind it, what is the chance of opening the prize-winning door?	
b. When reaching into a dark closet with five pairs of shoes, what is the chance of pulling out a left shoe?	
c. When prompted to randomly select a crayon from a group of red, blue, and yellow crayons, what is the chance you choose a purple crayon?	
d. When rolling a fair six-sided die, what is the chance of landing on 1?	
e. When flipping a coin, what is the chance of landing on either heads or tails?	

2. Plot and label a point to represent the percent chance of each event happening from the previous question. Use the letters labeling each event.

The **probability** of an event is a measure of the likelihood the event will occur. Probabilities are expressed using numbers from 0 to 1 as a **decimal**, **percentage**, or **fraction**.

3. Consider the event from question 1 that is least likely to occur.

   a. Which event is the least likely to happen?

   b. Describe, using sentences, the likelihood of this event happening.

   c. Discuss with your partner whether or not it is possible to have a probability less than this.

4. Consider the event from question 1 that is most likely to occur.

   a. Which event is the most likely to occur?

   b. Describe, in sentences, the likelihood of this event happening.

   c. Discuss with your partner whether or not it is possible to have a probability greater than this.

5. Which event has the same likelihood of happening or not happening?

15.3.3 Bring It Together: Representing Probabilities

1. Complete the statements.

   The probability of an event can be any number between 0 and 1. A probability of 0 means it is
   - ○ certain
   - ○ impossible

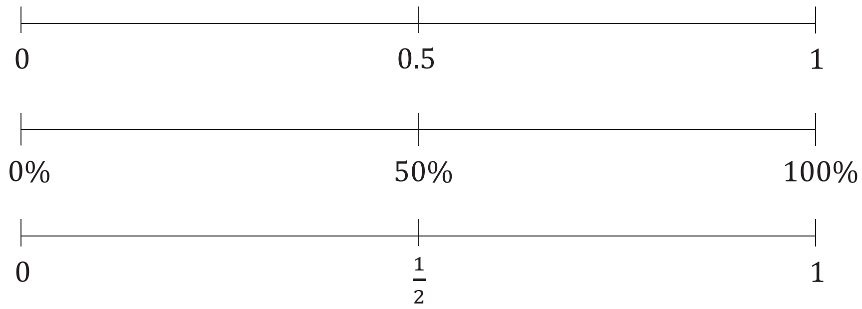
   for the event to occur. A probability of 1 means it is
   - ○ certain
   - ○ impossible

   the event will occur. The closer a probability is to 1, the
   - ○ more
   - ○ less

   likely it is to occur, while the closer it is to 0, the
   - ○ more
   - ○ less

   likely it is to occur. A probability of 0.5 means the event is
   - ○ likely.
   - ○ unlikely.
   - ○ neither likely nor unlikely.

Probabilities can be represented by decimals, percentages, or fractions with equivalent values between 0 and 1.



15.3.4 Guided Instruction: Expressing Probabilities of Events

The **probability** of an event occurring, notated as $P(event)$, is determined by dividing the number of outcomes in the event by the possible outcomes in the sample space.

$$P(event) = \frac{\text{number of outcomes in the event}}{\text{total possible outcomes}}$$

Harper is playing a game where she draws a lettered tile out of a bag without looking.

The notation $P(\text{A})$ in this context means "the probability of pulling out a tile with the letter A on it."

1. Find the probability of each event, represented as a fraction.

   a. $P(\text{A})$

   b. $P(\text{B})$

   c. $P(\text{M})$

   d. $P(\text{vowel})$

   e. $P(\text{letter})$

2. Consider the probability of pulling a tile with a consonant on it.

   a. Complete the table to find the fraction, decimal, and percentage equivalents of the probability.

Event	Fraction	Decimal	Percent
$P(\text{consonant})$			

The probability of this event is a repeating decimal. When this occurs, the decimal or percentage representation is typically rounded.

   b. What is $P(\text{consonant})$ represented as a decimal rounded to the nearest hundredth?

   c. What is $P(\text{consonant})$ represented as a percentage rounded to the nearest tenth?

15.3.5 Try It: Expressing Probabilities of Events in Different Forms

1. The probability of rolling a 6 on a standard six-sided die is 16.67%, rounded to the nearest hundredth. Maisie says that this probability is greater than an event whose probability is 0.6 because $16.67 > 0.6$.

   Explain whether you agree or disagree with Maisie's conclusion. Support your argument by showing your work.

2. The probabilities of five events are given.

A	B	C	D	E
40%	0.25	$\frac{5}{6}$	42.9%	$\frac{1}{5}$

   a. Order the values of the probabilities from least to greatest.

   b. Match the probabilities with the events described in the table.

Event	Probability
 Landing on green on the spinner shown	
Pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles	
Landing on a factor of 10 when rolling a fair ten-sided die	
Picking a card with a vowel on it from a set of cards with each letter in the word FLORIDA on it	
Landing on a number less than 6 on a standard six-sided die	

   c. Explain which event is the least likely to occur based on the probabilities.

15.3.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can express the probability of a chance event using a fraction, percentage, or decimal.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can compare probabilities of chance events.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Unit 15, Lesson 4: Comparing Probabilities

Benchmark

MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.

Learning Targets

- ☐ I can interpret the likelihood of a chance event occurring using probability.
- ☐ I can compare probabilities of chance events.

15.4.1 Warm-Up: Statisticians Careers

Statisticians turn data into understandable and useful information. The government, health care industry, sports broadcasting, and many businesses have statisticians who analyze data to predict trends. Statisticians help businesses and industries with decision-making based on analysis.

1. What are some real-world examples where you have seen statistics used?

2. Consider an industry you are interested in working in someday.

   a. What industry are you considering?

   b. What is at least one way you could see trends in data and analysis of statistics used in this industry?

15.4.2 Activity: Likelihood of Chance Events

Work with your partner or group to order the events on the cards from least likely to most likely.

After ordering the events, work with your partner or group to complete the following using the events from the card sort.

1. Explain which event(s), if any, can be described as "impossible."

2. Explain which event(s), if any, can be described as "certain."

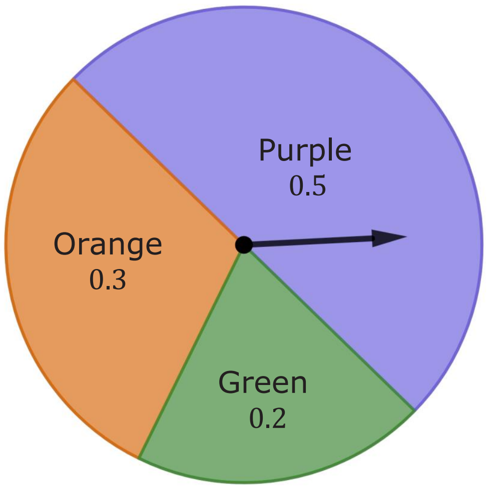
3. Explain which event(s), if any, can be described as "equally likely as not."

4. Express the likelihood of a randomly chosen word in a novel beginning with the letter T, as a percentage.

5. Express the likelihood of a randomly selected person being left-handed, as a decimal.

15.4.3 Guided Practice: Interpreting and Comparing Probabilities

1. A spinner is shown with the probabilities of landing on each space given as a decimal.



Use the spinner to complete the table by identifying the missing events, probabilities, and descriptions of likelihood.

Event	Probability	Likelihood
$P($______$)$	50%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain
$P(\text{orange})$	30%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain
$P(\text{yellow})$	____%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain
$P($______$)$	20%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain
$P(\text{purple or orange})$	80%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain
$P(\text{color})$	____%	○ impossible ○ unlikely ○ equally likely as not ○ likely ○ certain

2. For each scenario, circle the event that is more likely. If the events are equally likely, circle both.

   a. Alex and Elvis play baseball. Based on their statistics, Alex's probability of hitting a home run is 0.27. The probability of Elvis hitting a home run is 14%. Which choice is more likely on their next at bat?

Alex hits a home run.	or	Elvis hits a home run.

   b. A meteorologist states there is a 50% chance for rain tomorrow. Which choice is more likely to happen?

It will rain tomorrow.	or	It will not rain tomorrow.

   c. Shawn is calling the coin toss before his football game. Which choice is more likely to occur?

The coin will land on its edge.	or	The coin will land on either heads or tails.

15.4.4 Your Turn!

1. The probability that Lawrence makes a free throw when playing basketball is 60%. The likelihood that he makes a 3-point shot is 0.345.

   Explain which event is more likely, Lawrence making a free throw or making a 3-point shot.

2. The probabilities of five events are shown.

   60% 8 out of 10 0.37 20% $\frac{5}{6}$

   List the probabilities in order from least to greatest.

15.4.5 Wrap-Up: Ticket Out the Door

1. There are 25 prime numbers between 1 and 100. There are 46 prime numbers between 1 and 200. Two events are shown.

Event A	Event B
A computer produces a random number between 1 and 100 that is prime.	A computer produces a random number between 1 and 200 that is prime.

Ava said Event B is more likely to occur because there are 46 possible outcomes in the event compared to only 25 in Event A. Oscar said Event A is more likely because $P(\text{A}) = 0.25$ and $P(\text{B}) = 0.23$.

a. Explain who is correct.

b. Write a short note to the student who was incorrect explaining the error they made.

Unit 15, Lesson 5: Experimental and Theoretical Probability

Benchmarks

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Targets

- ☐ I can perform a simulation to determine an experimental probability.
- ☐ I can find the theoretical probability of an event.
- ☐ I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.

15.5.1 Warm-Up: Marbles

1. Fill in the blanks in the statements below so the probability of drawing a red marble from either bag is the same. Use any number 1 through 9 at most one time each.

   There are ______ red marbles and ______ blue marbles in Bag A. There are ______ red marbles and ______ green marbles in Bag B.

15.5.2 Exploration: Playing the Long Game

Samaria plays a game with a fair six-sided die in which she only wins if she rolls a 1 or a 2.

1. List the outcomes in the sample space for rolling the fair die.

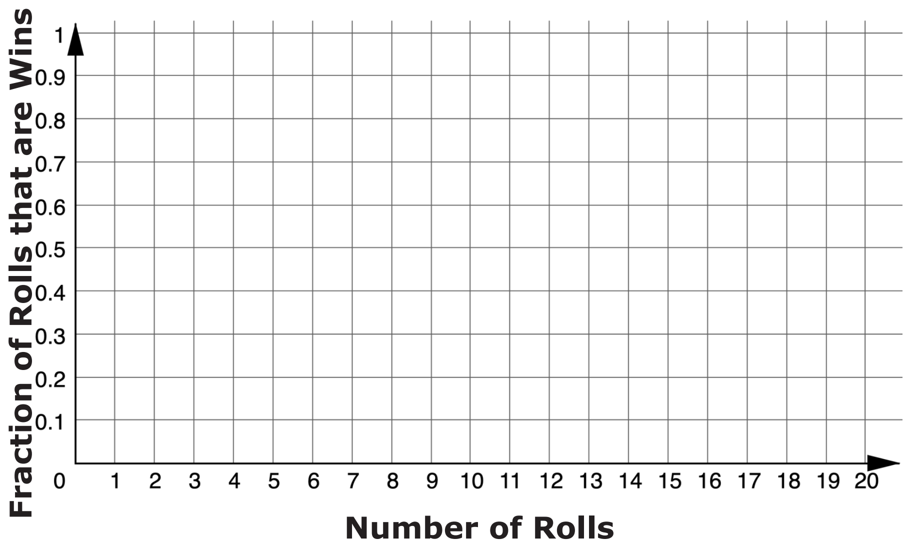
2. Discuss the probability of Samaria winning the game with your partner or group. Then, complete the statement.

   The probability Samaria wins the game, represented as a fraction, is _____ because there are _____ possible winning outcomes out of _____ total possible outcomes from rolling the die.

3. If Samaria is given the option to flip a coin and win if it comes up heads, explain whether or not that would be a better option for her to win using probability.

4. With your partner or group, follow these instructions 10 times to create the graph.
   - One person rolls the number cube. Everyone records the outcome.
   - Calculate the fraction of all rolls that are wins for Samaria so far. Approximate the fraction with a decimal value rounded to the hundredths place using a calculator. Record both the fraction and the decimal in the last column of the table.
   - On the graph, plot the number of rolls and the fraction of rolls that were wins.
   - Pass the number cube to the next person in the group.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			



5. With your partner or group, discuss what you notice is happening with the points on the graph.

   a. After 10 rolls, what fraction of the total rolls were a win?

   b. How close is this fraction to the probability that Samaria will win?

6. Roll the number cube 10 more times. Record your results in this table and add them to the graph from earlier.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			

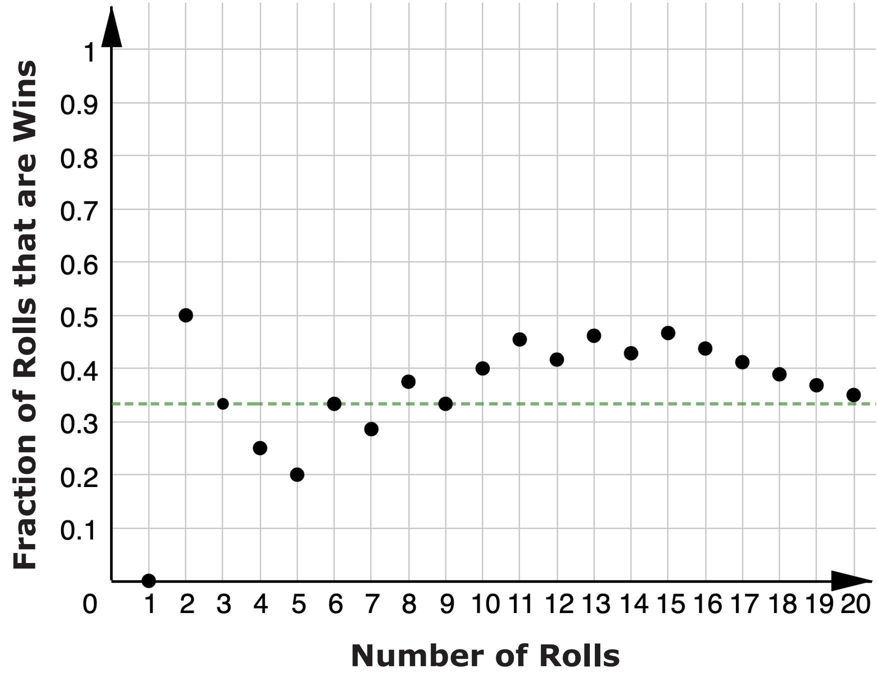
7. Observe any trends on the graph after 20 rolls.

   a. After 20 rolls, what fraction of the total rolls was a win?

   b. How close is this fraction to the probability that Samaria will win?

15.5.3 Bring It Together: Playing the Long Game

A sample graph from a group who performed the same simple experiment in the previous activity is shown.



1. What does the green dotted line represent on the graph?

2. Consider the data from all of the groups performing the experiment.

   a. Complete the table below with your class by adding the overall data from each group. The first row has already been completed using the sample data above.

Group	**Total Number of Wins for Samaria**	**Fraction of Games Played that are Wins**
Sample Data	7	$\frac{7}{20} = 0.35$

   b. Write the fraction that represents the total number of wins out of the total number of games played by all groups.

   c. How close is this fraction to the probability that Samaria will win?

The probability for an event represents the proportion of the time it is expected for the event to occur in the long run, over many trials. It does not mean that is exactly how many times the event will actually occur.

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Consider the terms defined below and how they relate to the previous Exploration.

A ***theoretical probability*** is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

An ***experimental probability*** is a number between 0 and 1 that shows the ratio of the number of times an event occurs to the total number of trials or times the activity is performed.

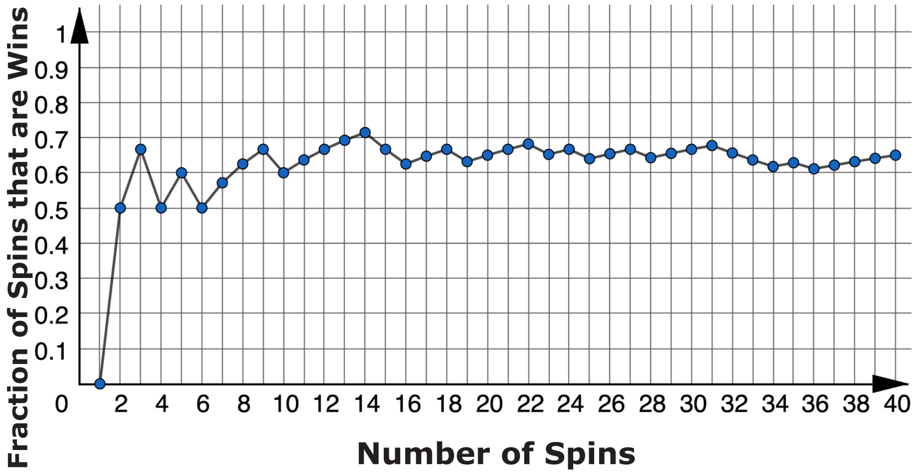
1. Use the definitions to complete the statements.

In the Exploration, the [○ experimental ○ theoretical] probability of Samaria winning the game was $\frac{1}{3}$ because there were [○ 2 ○ 3 ○ 6] possible outcomes out of [○ 2 ○ 3 ○ 6] in the sample space that won. By perfoming the activity, each group determined [○ an experimental ○ a theoretical] probability for the experiment.

15.5.5 Try It: Experimental and Theoretical Probability

1. Deepa wants to know if flipping a quarter really does have a probability of $\frac{1}{2}$ of landing heads-up, so she flips a quarter 10 times. It lands heads-up 3 times and tails-up 7 times. Explain whether or not she has proven that the probability of landing heads up is not $\frac{1}{2}$.

2. A spinner is spun 40 times for a game. The graph shows the fraction of games that are wins.



a. Estimate the probability of a spin winning this game based on the graph.

b. What type of probability is your estimate?

15.5.6 Wrap-Up: Cool Down

1. Your friend was absent today and will need the notes from class. Write a quick note to help them understand the difference between the theoretical and experimental probability of a simple experiment.

Unit 15, Lesson 6: Simulating Small and Large Numbers of Trials

Benchmark

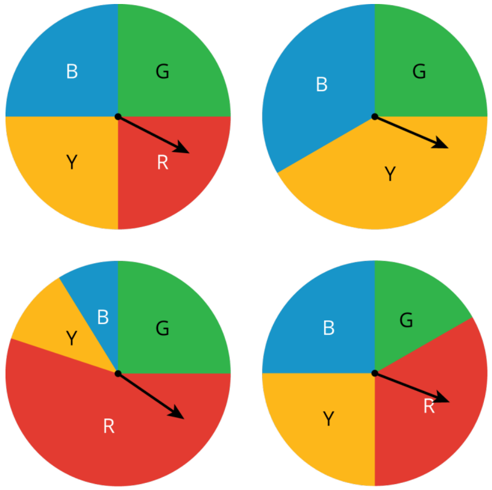
MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

Learning Targets

- ☐ I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.
- ☐ I can use results from computer-simulated events to informally define the Law of Large Numbers.

15.6.1 Warm-Up: Which One Doesn't Belong?

1. Four spinners are shown.



a. Which spinner doesn't belong?

b. Explain your reasoning.

15.6.2 Exploration: Comparing Experimental Results to Expected Results

Work with your partner to complete the following.

1. When performing the simple experiment of flipping a coin over repeated trials, imagine each of the resulting situations listed in the table.

   a. Complete the table to describe the results.

Situation	Is the result surprising?		Is the result possible?	
	Yes	No	Yes	No
You flip the coin once, and it lands heads-up.	○	○	○	○
You flip the coin twice, and it lands heads-up both times.	○	○	○	○
You flip the coin 100 times, and it lands heads-up all 100 times.	○	○	○	○

   b. If you flip the coin 100 times, explain how many times you would expect the coin to land heads-up.

   c. With your partner, discuss some other results that would not be surprising if you flip the coin 100 times.

2. A spinner is shown.

a. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Red	Purple	Orange	Blue	Yellow	Green
Theoretical Probability						

b. Sara used a computer-simulated spinner to perform different numbers of trials with the spinner. The results show the percentage of times it landed on each color in each simulation.

Circle the experimental probabilities that are more than 1% different from the theoretical probability.

Color	Red	Purple	Orange	Blue	Yellow	Green
Simulation A (6 spins)	0%	16.7%	16.7%	33.3%	0%	33.3%
Simulation B (60 spins)	16.7%	15%	20%	13.3%	10%	25%
Simulation C (600 spins)	17.7%	19%	18%	15.8%	14.5%	15%
Simulation D (6000 spins)	16.9%	16%	17.1%	16.6%	16.7%	17.1%

c. Discuss with your partner what you notice when comparing the experimental probabilities to the theoretical probabilities.

d. In which simulation(s) were the experimental probabilities the most different from the theoretical probabilities?

e. In which simulation(s) were the experimental probabilities the most similar to the theoretical probabilities?

f. Imagine Sara used the computer-simulated spinner to perform $600{,}000$ spins. Describe how you think the experimental probabilities of landing on each color would compare to the theoretical probabilities.

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results

A ***simulation*** is an approximate imitation of a statistical experiment, often done with a computer program to examine the statistics of a large quantity of trials.

1. Complete the statement.

   As the number of trials in a simulation of a simple experiment increases, the [○ experimental / ○ theoretical] probability gets [○ closer to / ○ farther from] the [○ experimental / ○ theoretical] probability.

2. Imagine flipping a coin 3 times, and it has come up heads once. The experimental probability of landing on heads after 3 trials is $\frac{1}{3}$. Discuss with your partner whether flipping the coin one more time is guaranteed to result in landing on heads so the experimental probability, $\frac{2}{4}$, will equal the theoretical probability $\frac{1}{2}$.

When performing simple experiments, the experimental probabilities may differ from theoretical probabilities due to random variation. However, as the number of repetitions increases, experimental probabilities will typically better approximate the theoretical probabilities.

Simulations are useful when it is hard or time consuming to gather enough information to estimate the probability of some event.

3. Consider the situations described in the table.

Situation	Description
A	The local meteorologist in Ocala, FL says there is a 50% chance of rain tomorrow.
B	A science teacher at Hamilton County High School puts the names of all students in a bag and randomly selects which of 6 groups they will sit in for his new seating chart.
C	The U.S. Centers for Disease Control and Prevention (CDC) released data in 2016 showing that $\frac{1}{3}$ of all adults in the US do not get enough sleep on a regular basis.

   Explain which situation could be simulated using the spinner from the Exploration.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

1. Four situations are described.

Situation	Description
A	In a small lake, 25% of the fish are female. You capture a fish, record whether it is male or female, and toss the fish back into the lake. If you repeat this process 5 times, what is the probability that at least 3 of the 5 fish are female?
B	Elena makes about 80% of her free throws. Based on her past successes with free throws, what is the probability that she will make exactly 4 out of 5 free throws in her next basketball game?
C	On a game show, a contestant must pick one of three doors. In the first round, the winning door has a vacation. In the second round, the winning door has a car. What is the probability of winning a vacation and a car?
D	Your choir is singing in 4 concerts. You and one of your classmates both learned the solo. Before each concert, there is an equal chance the choir director will select you or the other student to sing the solo. What is the probability that you will be selected to sing the solo in exactly 3 of the 4 concerts?

Work with your partner to match each situation to a simulation on the next page.

Simulation	Situation
Toss a standard number cube 2 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 1 or 2 appeared both times to estimate the probability.	
Make a spinner with four equal sections labeled $1, 2, 3,$ and 4. Spin the spinner 5 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 4 appears 3 or more times to estimate the probability.	
Toss a fair coin 4 times and record the outcomes. Repeat this process many times, and find the proportion of the simulations in which exactly 3 heads appear to estimate the probability.	
Place 8 blue chips and 2 red chips in a bag. Shake the bag, select a chip, record its color, and then return the chip to the bag. Repeat the process 4 more times to obtain a simulated outcome. Then repeat this process many times and find the proportion of the simulations in which exactly 4 blues are selected to estimate the probability.	

15.6.5 Wrap-Up: Reflection

1. What is something that is now "squared away" in your mind about simulating the probability of events?

2. What are three "points" that are important about simulating the probability of events?

3. What is something that is still "circling" your brain about simulating the probability of events?

Unit 15, Lesson 7: Experimental and Theoretical Probabilities from Simulations

Benchmarks

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

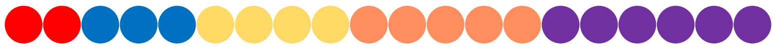
Learning Target

- [ ] I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.

15.7.1 Warm-Up: Visualizing Probabilities

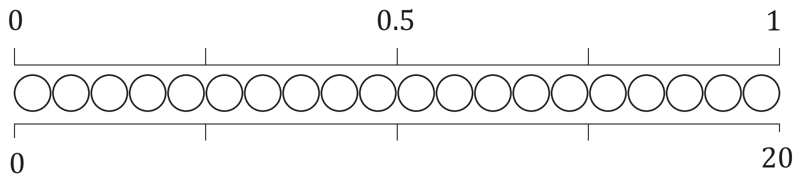
Ashley places 2 red marbles, 3 blue marbles, 4 yellow marbles, 5 orange marbles, and 6 purple marbles in a bag.

A representation of the marbles is shown.

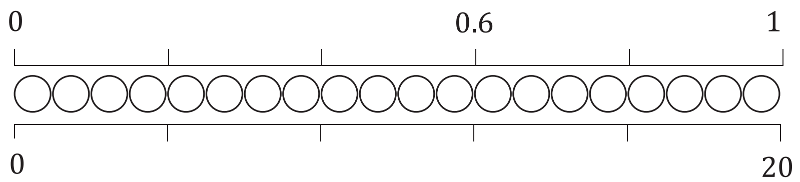


1. Complete the double number line diagrams below by writing values at each tick mark. The top line represents probabilities, and the bottom line represents the number of marbles in the bag.

   a.



   b.



2. Write a color on each blank to complete the statements.

   The probability of pulling a ________________ marble out of the bag without looking is 0.2. The probability of pulling a ________________ marble out of the bag without looking is 0.25.

15.7.2 Collaboration: Simulating Probabilities

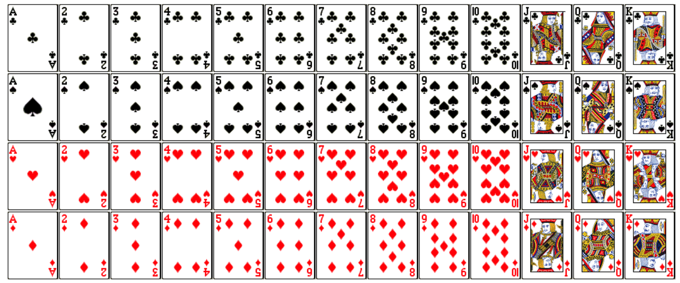
1. In the 2019-2020 NBA regular season, Miami Heat player Jimmy Butler made $\frac{1}{4}$ of the three-point shots he attempted. What was the probability of Jimmy Butler making a three-point shot in the 2019-2020 regular season, represented as a decimal?

2. Work with your partner to complete each statement to describe ways Jimmy's three-point shot probability could be simulated using different simple experiments.

   a. Create a spinner with _____ equal section(s), where _____ section(s) represent(s) Jimmy Butler making a three-point shot and the other(s) represent(s) misses.

   b. Place twelve equal-sized slips of paper with the numbers $1-12$ written on them in a bag. Papers with the numbers _______________ represent Jimmy Butler making a three-point shot and the others represent misses.

   c. Use a standard deck of 52 cards where ____________________________________ represent Jimmy Butler making a three-point shot and the other cards represent misses.



3. Suppose in the first post-season game that year, Jimmy Butler attempted 5 three-point shots. Choose one of the simulations from the previous question and work with your partner or group to simulate the number of three-point shots Jimmy made.

   a. Repeat the simulation 5 times and record your results in the table.

	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Fraction Made
Simulation 1						$\frac{\ \ }{5}$
Simulation 2						$\frac{\ \ }{5}$
Simulation 3						$\frac{\ \ }{5}$
Simulation 4						$\frac{\ \ }{5}$
Simulation 5						$\frac{\ \ }{5}$

   b. Based on your group's simulations, estimate the probability that Jimmy Butler made $\frac{2}{5}$ of the three-point shots he attempted in the game.

   c. Jimmy Butler actually made 3 out of the 5 three-point shots he attempted in the game. Explain how that compares to your group's simulation results.

4. In the 2019-2020 NBA regular season, Orlando Magic player Aaron Gordon made $\frac{2}{3}$ of free-throw shots attempted. Work with your partner to describe how each of the following could be used to simulate his attempted free-throw shots.

   a. A fair 6-sided die

   b. A standard 52-card deck

   c. Different colored marbles placed in a bag

15.7.3 Your Turn!

1. Suppose a fair six-sided die numbered 1 through 6 is rolled 12 times.

   a. Of those 12 rolls, how many do you expect will land on the number 3?

   _____ times out of 12 rolls

   b. Roll a standard six-sided die 12 times and record the number on the top of the cube when it lands.

Roll 1	Roll 2	Roll 3	Roll 4	Roll 5	Roll 6	Roll 7	Roll 8	Roll 9	Roll 10	Roll 11	Roll 12

   c. What is the theoretical probability of rolling a 3?

   d. Describe how the theoretical probability compares to your experimental probability.

   e. Review your results and your partner's results. Explain why the results are not the same.

2. Consider the events of rolling a fair 10-sided die and getting an odd number or flipping a coin and having it land tails-up. Explain which event is more likely.

3. Noelle picks a card at random from a stack that has cards numbered 11 through 25.

   a. Determine the sample space.

   b. Find each probability.

   $P(\text{even}) =$

   $P(\text{multiple of } 5) =$

   $P(\text{two digit number}) =$

   $P(\text{number less than } 10) =$

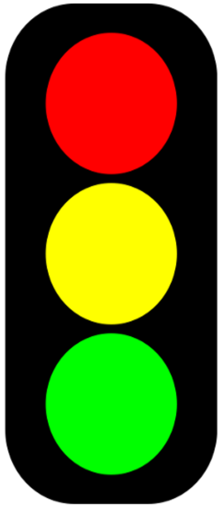
   c. Noelle used a simulator to run the experiment 300 times. She selected an odd-numbered card 154 times. Compare the experimental probability to the theoretical probability of selecting an odd-numbered card.

15.7.4 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about probability, including those listed below.

- Determine the sample space for a simple experiment
- Express probabilities as fractions, decimals, and percentages
- Interpret and compare the probability of chance events
- Find the theoretical probability of an event related to a simple experiment
- Compare experimental probability from simulations to theoretical probability

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck on how to …

I'm starting to get the hang of …

I feel confident and ready to go with …

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